

9.2 Measuring Area



Figure 9.5 A painter uses an extension roller to paint a wall. (credit: "Paint Rollers are effective" by WILLPOWER STUDIOS/ Flickr, CC BY 2.0)

Learning Objectives

After completing this section, you should be able to:

1. Identify reasonable values for area applications.
2. Convert units of measures of area.
3. Solve application problems involving area.

Area is the size of a surface. It could be a piece of land, a rug, a wall, or any other two-dimensional surface with attributes that can be measured in the metric unit for distance-meters. Determining the area of a surface is important to many everyday activities. For example, when purchasing paint, you'll need to know how many **square units** of surface area need to be painted to determine how much paint to buy.

Square units indicate that two measures in the same units have been multiplied together. For example, to find the area of a rectangle, multiply the length units and the width units to determine the area in square units.

$$1 \text{ cm} \times 1 \text{ cm} = 1 \text{ cm}^2$$

Note that to accurately calculate area, each of the measures being multiplied must be of the same units. For example, to find an area in square centimeters, both length measures (length and width) must be in centimeters.

FORMULA

The formula used to determine area depends on the shape of that surface. Here we will limit our discussions to the area of rectangular-shaped objects like the one in Figure 9.6. Given this limitation, the basic formula for area is:

$$\text{Area} = \text{length } (l) \times \text{width } (w)$$

or

$$A = l \times w$$

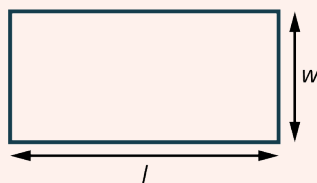


Figure 9.6 Rectangle with Length (l) and Width (w) Labeled

Reasonable Values for Area

Because area is determined by multiplying two lengths, the magnitude of difference between different square units is exponential. In other words, while a meter is 100 times greater in length than a centimeter, a square meter (m^2) is 100×100 , or 10,000 times greater in area than a square centimeter (cm^2). The relationships between benchmark metric area units are shown in the following table.

Units	Relationship	Conversion Rate
km^2 to m^2	$\text{km} \times \text{km} = \text{km}^2$ $1 \text{ km} = 1,000 \text{ m}$ $1,000 \text{ m} \times 1,000 \text{ m} = 1,000,000 \text{ m}^2$	$1 \text{ km}^2 = 1,000,000 \text{ m}^2$
m^2 to cm^2	$\text{m} \times \text{m} = \text{m}^2$ $1 \text{ m} = 100 \text{ cm}$ $100 \text{ cm} \times 100 \text{ cm} = 10,000 \text{ cm}^2$	$1 \text{ m}^2 = 10,000 \text{ cm}^2$
cm^2 to mm^2	$\text{cm} \times \text{cm} = \text{cm}^2$ $1 \text{ cm} = 10 \text{ mm}$ $10 \text{ mm} \times 10 \text{ mm} = 100 \text{ mm}^2$	$1 \text{ cm}^2 = 100 \text{ mm}^2$

An essential understanding of metric area is to identify reasonable values for area. When testing for reasonableness you should assess both the unit and the unit value. Only by examining both can you determine whether the given area is reasonable for the situation.

EXAMPLE 9.10

Determining Reasonable Units for Area

Which unit of measure is most reasonable to describe the area of a sheet of paper: km^2 , cm^2 , or mm^2 ?

Solution

In the U.S. Customary System of Measurement, the length and width of paper is usually measured in inches. In the metric system centimeters are used for measures usually expressed in inches. Thus, the most reasonable unit of measure to describe the area of a sheet of paper is square centimeters. Square kilometers is too large a unit and square millimeters is too small a unit.

YOUR TURN 9.10

1. Which unit of measure is most reasonable to describe the area of a forest: km^2 , cm^2 , or mm^2 ?

EXAMPLE 9.11

Determining Reasonable Values for Area

You want to paint your bedroom walls. Which represents a reasonable value for the area of the walls: 100 cm^2 , 100 m^2 , or 100 km^2 ?

Solution

An area of 100 cm^2 is equivalent to a surface of $10 \text{ cm} \times 10 \text{ cm}$, which is much too small for the walls of a bedroom. An area of 100 km^2 is equivalent to a surface of $10 \text{ km} \times 10 \text{ km}$, which is much too large for the walls of a bedroom. So, a reasonable value for the area of the walls is 100 m^2 .

> YOUR TURN 9.11

1. Which represents a reasonable value for the area of the top of a kitchen table: $1,800 \text{ mm}^2$, $1,800 \text{ cm}^2$, or $1,800 \text{ m}^2$?

EXAMPLE 9.12**Explaining Reasonable Values for Area**

A landscaper is hired to resod a school's football field. After measuring the length and width of the field they determine that the area of the football field is $5,350 \text{ km}^2$. Does their calculation make sense? Explain your answer.

✓ Solution

No, kilometers are used to determine longer distances, such as the distance between two points when driving. A football field is less than 1 kilometer long, so a more reasonable unit of value would be m^2 . An area of $5,350 \text{ km}^2$ can be calculated using the dimensions 53.5 by 100, which are reasonable dimensions for the length and width of a football field. So, a more reasonable value for the area of the football field is $5,350 \text{ m}^2$.

> YOUR TURN 9.12

1. A crafter uses four letter-size sheets of paper to create a paper mosaic in the shape of a square. They decide they want to frame the paper mosaic, so they measure and determine that the area of the paper mosaic is $2,412.89 \text{ cm}^2$. Does their calculation make sense? Explain your answer.

Converting Units of Measures for Area

Just like converting units of measure for distance, you can convert units of measure for area. However, the conversion factor, or the number used to multiply or divide to convert from one area unit to another, is not the same as the conversion factor for metric distance units. Recall that the conversion factor for area is exponentially relative to the conversion factor for distance. The most frequently used conversion factors are shown in [Figure 9.7](#).

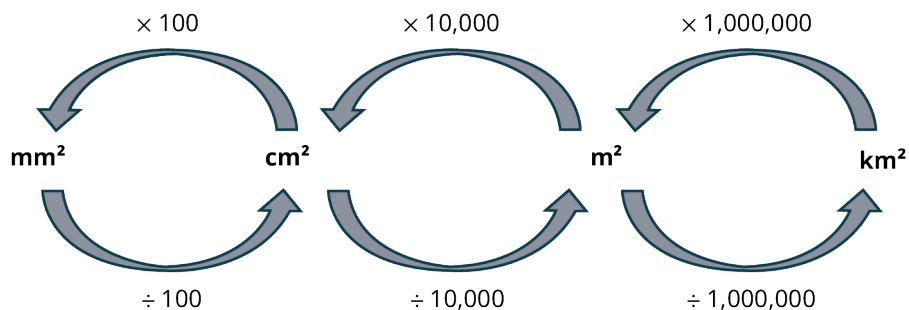


Figure 9.7 Common Conversion Factors for Metric Area Units

▶ VIDEO

[Converting Metric Units of Area \(https://openstax.org/r/Metric_Units_of_Area\)](https://openstax.org/r/Metric_Units_of_Area)

EXAMPLE 9.13**Converting Units of Measure for Area Using Division**

A plot of land has an area of $237,500,000 \text{ m}^2$. What is the area in square kilometers?

✓ Solution

Use division to convert from a smaller metric area unit to a larger metric area unit. To convert from m^2 to km^2 , divide the value of the area by 1,000,000.

$$\frac{237,500,000}{1,000,000} = 237.5$$

The plot of land has an area of 237.5 km².

YOUR TURN 9.13

1. A roll of butcher paper has an area of 1,532,900 cm². What is the area of the butcher paper in square meters?

EXAMPLE 9.14

Converting Units of Measure for Area Using Multiplication

A plot of land has an area of 0.004046 km². What is the area of the land in square meters?

Solution

Use multiplication to convert from a larger metric area unit to a smaller metric area unit. To convert from km² to m², multiply the value of the area by 1,000,000.

$$0.004046 \times 1,000,000 = 4,046$$

The plot of land has an area of 4,046 m².

YOUR TURN 9.14

1. A bolt of fabric has an area of 136.5 m². What is the area of the bolt of fabric in square centimeters?

EXAMPLE 9.15

Determining Area by Converting Units of Measure for Length First

A computer chip measures 10 mm by 15 mm. How many square centimeters is the computer chip?

Solution

Step 1: Convert the measures of the computer chip into centimeters

$$10 \text{ mm} = 1 \text{ cm}$$

$$15 \text{ mm} = 1.5 \text{ cm}$$

Step 2: Use the [area formula](#) to determine the area of the chip.

$$1 \times 1.5 = 1.5$$

The computer chip has an area of 1.5 cm².

YOUR TURN 9.15

1. A piece of fabric measures 100 cm by 106 cm. What is the area of the fabric in square meters?

Solving Application Problems Involving Area

While it may seem that solving area problems is as simple as multiplying two numbers, often determining area requires more complex calculations. For example, when measuring the area of surfaces, you may need to account for portions of the surface that are not relevant to your calculation.

EXAMPLE 9.16**Solving for the Area of Complex Surfaces**

One side of a commercial building is 12 meters long by 9 meters high. There is a rolling door on this side of the building that is 4 meters wide by 3 meters high. You want to refinish the side of the building, but not the door, with aluminum siding. How many square meters of aluminum siding are required to cover this side of the building?

 Solution

Step 1: Determine the area of the side of the building.

$$12 \text{ m} \times 9 \text{ m} = 106 \text{ m}^2$$

Step 2: Determine the area of the door.

$$4 \text{ m} \times 3 \text{ m} = 12 \text{ m}^2$$

Step 3: Subtract the area of the door from the area of the side of the building.

$$106 \text{ m}^2 - 12 \text{ m}^2 = 92 \text{ m}^2$$

So, you need to purchase 92 m^2 of aluminum siding.

**YOUR TURN 9.16**

1. You want to cover a garden with topsoil. The garden is 5 meters by 8 meters. There is a path in the middle of the garden that is 8 meters long and 0.75 meters wide. What is the area of the garden you need to cover with topsoil?

When calculating area, you must ensure that both distance measurements are expressed in terms of the same distance units. Sometimes you must convert one measurement before using the [area formula](#).

EXAMPLE 9.17**Solving for Area with Distance Measurements of Different Units**

A national park has a land area in the shape of a rectangle. The park measures 2.2 kilometers long by 1,250 meters wide. What is the area of the park in square kilometers?

 Solution

Step 1: Use a conversion fraction to convert the information given in meters to kilometers.

$$1,250 \text{ m} \times \frac{1 \text{ km}}{1,000 \text{ m}} = 1.25 \text{ km}$$

Step 2: Multiply to find the area.

$$2.2 \text{ km} \times 1.25 \text{ km} = 2.75 \text{ km}^2$$

The park has an area of 2.75 km^2 .

**YOUR TURN 9.17**

1. An Olympic pool measures 50 meters by 2,500 centimeters. What is the surface area of the pool in square

meters?

When calculating area, you may need to use multiple steps, such as converting units and subtracting areas that are not relevant.

EXAMPLE 9.18**Solving for Area Using Multiple Steps**

A kitchen floor has an area of 15 m^2 . The floor in the kitchen pantry is 100 cm by 200 cm . You want to tile the kitchen and pantry floors using the same tile. How many square meters of tile do you need to buy?

 Solution

Step 1: Determine the area of the pantry floor in square centimeters.

$$100 \text{ cm} \times 200 \text{ cm} = 20,000 \text{ cm}^2$$

Step 2: Divide the area in cm^2 by the conversion factor to determine the area in m^2 since the other measurement for the kitchen floor is in m^2 .

$$\frac{20,000}{10,000} = 2$$

The area of the kitchen pantry floor is 2 m^2 .

Step 3: Add the two areas of the pantry and the kitchen floors together.

$$15 \text{ m}^2 + 2 \text{ m}^2 = 17 \text{ m}^2$$

So, you need to buy 17 m^2 of tile.

**YOUR TURN 9.18**

1. Your bedroom floor has an area of 25 m^2 . The living room floor measures 600 cm by 750 cm . How many square meters of carpet do you need to buy to carpet the floors in both rooms?

 WHO KNEW?**The Origin of the Metric System**

The metric system is the official measurement system for every country in the world except the United States, Liberia, and Myanmar. But did you know it originated in France during the French Revolution in the late 18th century? At the time there were over 250,000 different units of weights and measures in use, often determined by local customs and economies. For example, land was often measured in days, referring to the amount of land a person could work in a day.

**VIDEO**

[Why the Metric System Matters \(https://openstax.org/r/Metric_System_Matters\)](https://openstax.org/r/Metric_System_Matters)

Check Your Understanding

For the following exercises, determine the most reasonable value for each area.

9. bedroom wall: 12 km^2 , 12 m^2 , 12 cm^2 , or 12 mm^2
10. city park: $1,200 \text{ km}^2$, $1,200 \text{ m}^2$, $1,200 \text{ cm}^2$, or $1,200 \text{ mm}^2$

11. kitchen table: 2.5 km^2 , 2.5 m^2 , 2.5 cm^2 , or 2.5 mm^2

For the following exercises, convert the given area to the units shown.

12. $20,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
 13. $5.7 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
 14. $217 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
15. A wall measures 4 m by 2 m. A doorway in the wall measures 0.5 m by 1.6 m. What is the area of the wall not taken by the door in square meters?



SECTION 9.2 EXERCISES

For the following exercises, determine the most reasonable value for each area.

1. kitchen floor: 16 km^2 , 16 m^2 , 16 cm^2 , or 16 mm^2
2. national Park: $1,000 \text{ km}^2$, $1,000 \text{ m}^2$, $1,000 \text{ cm}^2$, or $1,000 \text{ mm}^2$
3. classroom table: 5 km^2 , 5 m^2 , 5 cm^2 , or 5 mm^2
4. window: $9,000 \text{ km}^2$, $9,000 \text{ m}^2$, $9,000 \text{ cm}^2$, or $9,000 \text{ mm}^2$
5. paper napkin: $10,000 \text{ km}^2$, $10,000 \text{ m}^2$, $10,000 \text{ cm}^2$, or $10,000 \text{ mm}^2$
6. parking lot: $45,000 \text{ km}^2$, $45,000 \text{ m}^2$, $45,000 \text{ cm}^2$, or $45,000 \text{ mm}^2$

For the following exercises, convert the given area to the units shown.

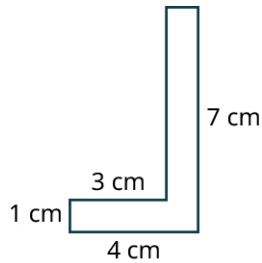
7. $700,000 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
8. $24 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
9. $985 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
10. $4 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$
11. $0.00005 \text{ km}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
12. $68,500,000 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
13. $0.005 \text{ m}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
14. $7,800 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ m}^2$
15. $34.5 \text{ m}^2 = \underline{\hspace{2cm}} \text{ cm}^2$
16. $0.05 \text{ km}^2 = \underline{\hspace{2cm}} \text{ m}^2$
17. $0.073 \text{ m}^2 = \underline{\hspace{2cm}} \text{ mm}^2$
18. $27,750 \text{ mm}^2 = \underline{\hspace{2cm}} \text{ m}^2$

For the following exercises, determine the area.

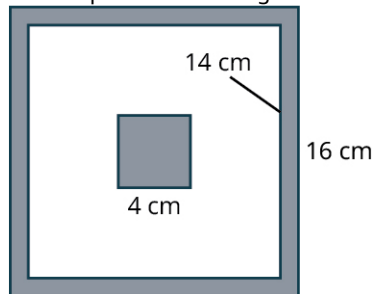
19. $l = 300 \text{ cm}$, $w = 4 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ m}^2$
 20. $l = 3,500 \text{ mm}$, $w = 0.5 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 21. $l = 5.5 \text{ km}$, $w = 2,750 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
 22. $l = 2,400 \text{ cm}$, $w = 3.2 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ m}^2$
 23. $l = 7 \text{ cm}$, $w = 400 \text{ mm}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 24. $l = 2,300 \text{ m}$, $w = 4 \text{ km}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
 25. $l = 4.2 \text{ cm}$, $w = 320 \text{ mm}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 26. $l = 5,350 \text{ mm}$, $w = 0.5 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ cm}^2$
 27. $l = 1.8 \text{ km}$, $w = 2,300 \text{ m}$
 $A = \underline{\hspace{2cm}} \text{ km}^2$
28. A notebook is 200 mm by 300 mm. A sticker on the notebook cover measures 2 cm by 2 cm. How many square centimeters of the notebook cover is still visible?
29. A bedroom wall is 4 m by 2.5 m. A window on the wall is 1 m by 2 m. How much wallpaper is needed to cover the wall?
30. A wall is 5 m by 2.5 m. A picture hangs on the wall that is 600 mm by 300 mm. How much of the wall, in m^2 , is not

covered by the picture?

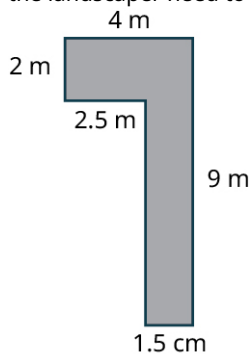
31. What is the area of the shape that is shown?



32. A quilter made a design using a small square, a medium square, and a large square. What is the area of the shaded parts of the design that is shown?



33. A landscaper makes a plan for a walkway in a backyard, as shown. How many square meters of patio brick does the landscaper need to cover the walkway?



34. A painter completed a portrait. The height of the portrait is 36 cm. The width is half as long as the height. What is the area of the portrait in square meters?
35. A room has an area of 137.5 m^2 . The length of the room is 11 m. What is the width of the room?
36. A wall is 4.5 m long by 3 m high. A can of paint will cover an area of 10 m^2 . How many cans of paint are needed if each wall needs 2 coats of paint?
37. A soccer field is 110 meters long and 75 meters wide. If the cost of artificial turf is \$30 per square meter, what is the cost of covering the soccer field with artificial turf?
38. A window is 150 centimeters wide and 90 centimeters high. If three times the area of the window is needed for curtain material, how much curtain material is needed in square meters?
39. A rectangular flower garden is 5.5 meters wide and 8.4 meters long. A path with a width of 1 meter is laid around the garden. What is the area of the path?
40. A dining room is 8 meters wide by 6 meters long. Wood flooring costs \$12.50 per square meter. How much will it cost to install wood flooring in the dining room?